David C. Lay: Linear Algebra and it's Applications

Exercise 2.1

28) If u and v are in $\mathbb{R}^n$, how are $u^T v$ and $v^T u$ related? How are uv^T and vu^T related?

$\boxed{\text{Solution:}}$ $(u^T v)^T = v^T (u^T)^T$ (Theorem 3(d))
$= v^T u$ (Theorem 3(a))

$(uv^T)^T = (v^T)^T u^T$ (Theorem 3(d))
$= vu^T$ (Theorem 3(a))

29) Prove Theorem 2(b) and 2(c). Use the row-column rule. The (i,j)-entry in $A(B+C)$ can be written as
$a_{i1}(b_{1j}+c_{1j}) + \dots\dots + a_{in}(b_{nj}+c_{nj})$ or $\boxed{\sum_{k=1}^{n} a_{ik}(b_{kj}+c_{kj})}$

$\boxed{\text{Solution:}}$ Theorem 2(b): $A(B+C) = AB+AC$

The (i,j)th entry in $A(B+C)$ is $\sum_{k=1}^{n} a_{ik}(b_{kj}+c_{kj})$
$= \sum_{k=1}^{n}(a_{ik}\cdot b_{kj} + a_{ik}\cdot c_{kj}) = \sum_{k=1}^{n} a_{ik}\cdot b_{kj} + \sum_{k=1}^{n} a_{ik}\cdot c_{kj}$

By the Row-Column rule, the (i,j)th entry of AB is
$a_{i1}\cdot b_{1j} + a_{i2}\cdot b_{2j} + \dots\dots + a_{in}\cdot b_{nj} = \boxed{\sum_{k=1}^{n} a_{ik}\cdot b_{kj}}$

By the Row-Column rule, the (i,j)th entry of AC is
$a_{i1}\cdot c_{1j} + a_{i2}\cdot c_{2j} + \dots\dots + a_{in}\cdot c_{nj} = \boxed{\sum_{k=1}^{n} a_{ik}\cdot c_{kj}}$

$\therefore$ The (i,j)th entry of $A(B+C)$ is the sum of the (i,j)th entry of AB and AC.

$\boxed{\therefore A(B+C) = AB+AC}$ □

$\boxed{(B+C)A = BA+CA} \rightarrow$ Theorem 2(c)

The (i,j)th entry in $(B+C)A$ by the Row-Column Rule is
$(b_{i1}+c_{i1})\cdot a_{1j} + (b_{i2}+c_{i2})\cdot a_{2j} + \dots\dots + (b_{in}+c_{in})\cdot a_{nj}$
$= \sum_{k=1}^{n}(b_{ik}+c_{ik})\cdot a_{kj}$